Opening Remarks

Alessandro Erba

Imperial College London

UNIVERSITÀ DI TORINO

Summer School on

Ab initio Modelling in Solid State Chemistry

First Principles Modelling in Solid State with CRYSTAL

MSSC2025

Department of Chemistry
Thomas Young Centre

@ Imperial College London

Department of Chemistry
Theoretical Chemistry Group

@ Università di Torino

Imperial College London

UNIVERSITÀ DI TORINO

MSSC is an established series of Summer Schools (**38 and counting**) running since the year **2000**, hosted by several institutions around the globe:

Torino, **Italy**
London, **UK**

Regensburg, **Germany**
Pau, **France**
La Plata, **Argentina**
Spokane, **USA**
Minneapolis, **USA**
Barcelona, **Spain**
Natal, **Brazil**
Jhansi, **India**

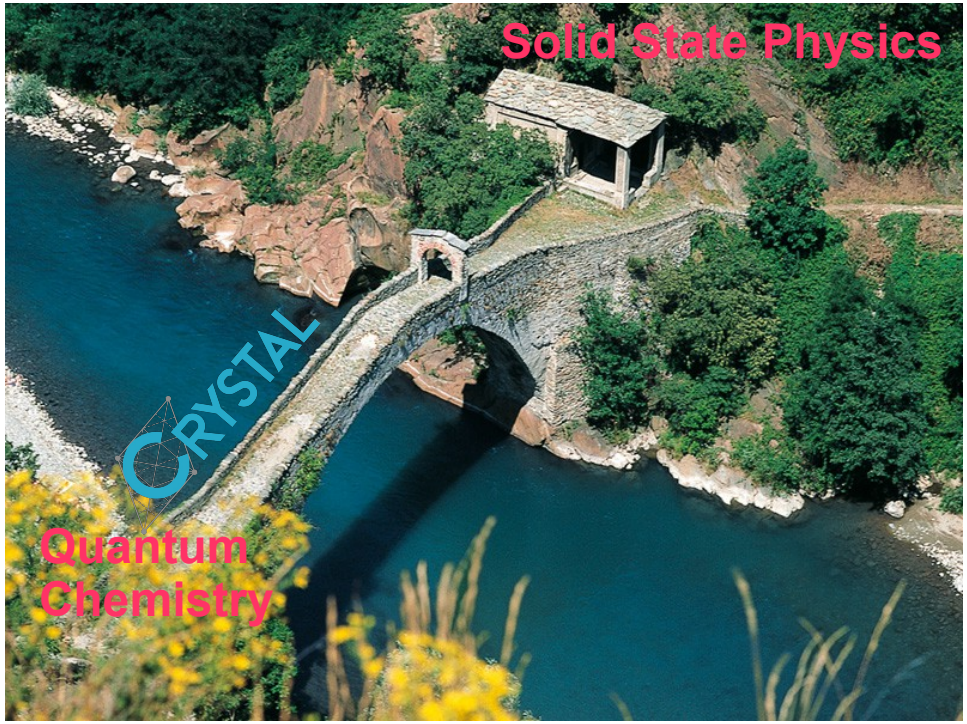
MSSC is a **mix** of **lectures** from experts in the field of computational solid state chemistry and physics and **hands-on sessions**, based on the use of the CRYSTAL software.

CRYSTAL

Imperial College London

UNIVERSITÀ DI TORINO

MSSC aims at bridging molecular quantum chemistry and computational solid state physics:



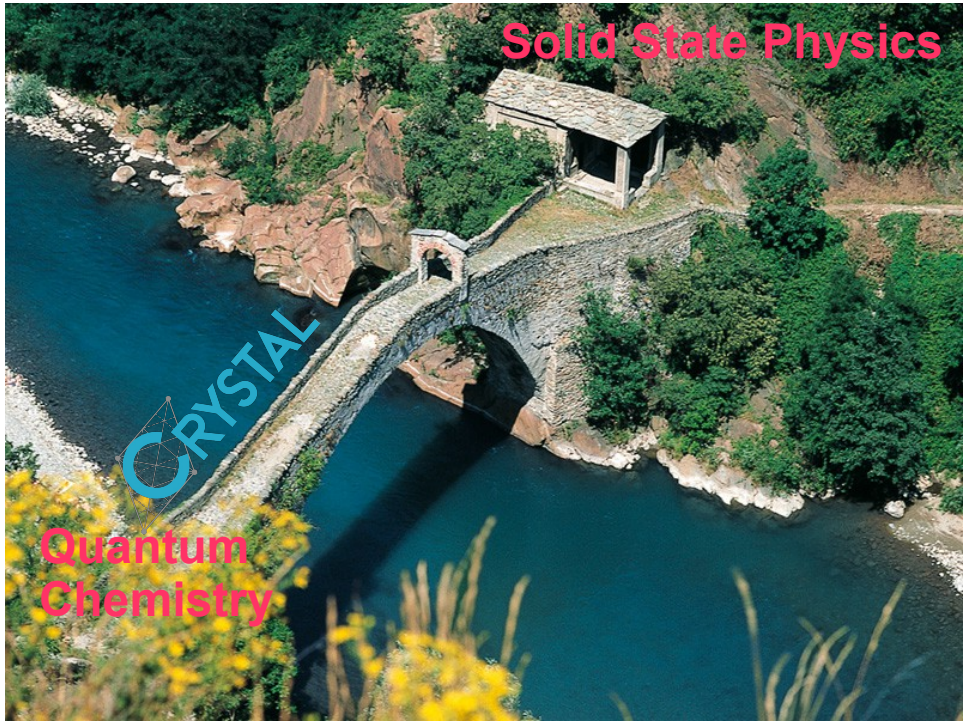
Ponte del Diavolo (Devil's Bridge), Lanzo Torinese

CRYSTAL is a first principles electronic structure package

Imperial College London

UNIVERSITÀ DI TORINO

MSSC aims at bridging molecular quantum chemistry and computational solid state physics:



Ponte del Diavolo (Devil's Bridge), Lanzo Torinese

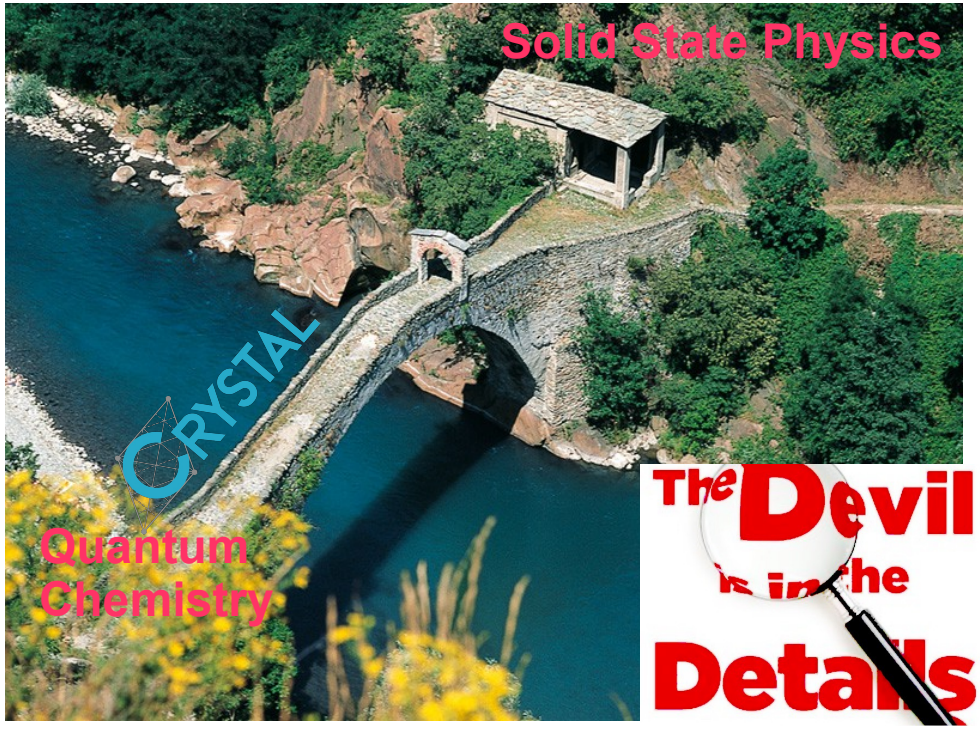
QC	SSP
(Extended) Huckel	Tight Binding
Coupled-Perturbed HF/KS (CPHF/KS)	Density Functional Perturbation Theory (DFPT)
Dynamical CPKS	Time-Dependent Sternheimer Equation
Vibrational Self-Consistent Field (VSCF)	Self-Consistent Harmonic Approximation, Self-Consistent Phonons (SCHA, SCP)
VPT, VCI	

CRYSTAL is a first principles electronic structure package

Imperial College London

UNIVERSITÀ DI TORINO

MSSC aims at bridging molecular quantum chemistry and computational solid state physics:



Ponte del Diavolo (Devil's Bridge), Lanzo Torinese

QC	SSP
(Extended) Huckel	Tight Binding
Coupled-Perturbed HF/KS (CPHF/KS)	Density Functional Perturbation Theory (DFPT)
Dynamical CPKS	Time-Dependent Sternheimer Equation
Vibrational Self-Consistent Field (VSCF)	Self-Consistent Harmonic Approximation, Self-Consistent Phonons (SCHA, SCP)
VPT, VCI	

CRYSTAL is a first principles electronic structure package, with a distinctive feature

- Use of atom-centered local basis functions versus plane-waves;

The use of an atom-centered local basis set (**CO-LCAO**) has many interesting implications:

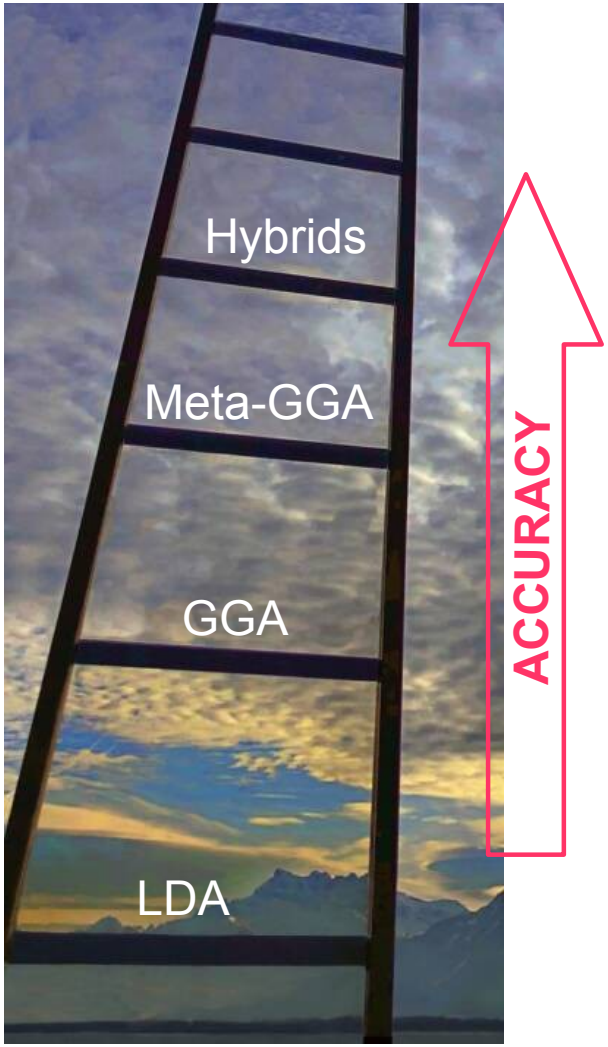
- It is easier to transfer "molecular" quantum-chemical algorithms to solids;
- It is more natural to exploit point-symmetry;
- It is possible to model low-dimensionality systems (0D, 1D, 2D) with no need to artificially replicate it along "non periodic" directions;
- It is natural to decompose various electronic properties (density-of-states, electron density, etc.) in atomic orbital contributions;
- It is possible to run very efficiently Meta-GGA calculations;
- Through overlap-based pre-screening, it is possible to compute very efficiently the exact non-local Fock exchange (cost of a hybrid calculation 2-6X that of a GGA calculation);

Imperial College London

UNIVERSITÀ DI TORINO

The use of an atom-centered local basis set (**CO-LCAO**) has many interesting implications:

- It is easier to transfer "molecular" quantum-chemical algorithms to solids;
- It is more natural to exploit point-symmetry;
- It is possible to model low-dimensionality systems (0D, 1D, 2D) with no need to artificially replicate it along "non periodic" directions;
- It is natural to decompose various electronic properties (density-of-states, electron density, etc.) in atomic orbital contributions;
- It is possible to run very efficiently Meta-GGA calculations;
- Through overlap-based pre-screening, it is possible to compute very efficiently the exact non-local Fock exchange (cost of a hybrid calculation 2-6X that of a GGA calculation);



Jacob's Ladder of DFT approximations

Imperial College London

UNIVERSITÀ DI TORINO

A couple of milestones in the development of exact Fock exchange in CRYSTAL:

i) The efficient implementation of the **Fock exchange** lattice series (in **1983**);

ii) The first ever periodic implementation of the **Hartree-Fock method** (publicly released in **1988**: CRYSTAL88);

iii) A first **mixed DFT/HF approach** for solids implemented in **1987** with an *a posteriori* correction to the HF total energy through the use of the Colle-Salvetti density functional for the correlation energy;

iv) In **1996**, shortly after 1993 Becke's original proposal, the implementation of global **hybrid density functional approximations** for solids (publicly released with CRYSTAL98);

Imperial College London

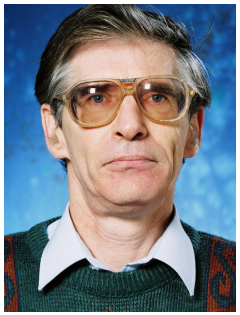
UNIVERSITÀ DI TORINO

The CRYSTAL project started in the **70s** in Torino, with a key collaboration with Victor R. Saunders ("Vic") from Daresbury, UK, in the early 80s.

Cesare Pisani

Roberto Dovesi

Carla Roetti



Vic R. Saunders

Imperial College London

UNIVERSITÀ DI TORINO

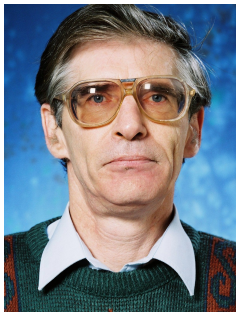
CRYSTAL88

The CRYSTAL project started in the **70s** in Torino, with a key collaboration with Victor R. Saunders ("Vic") from Daresbury, UK, in the early 80s.

Cesare Pisani

Roberto Dovesi

Carla Roetti



Vic R. Saunders

Imperial College London

UNIVERSITÀ DI TORINO

- CRYSTAL88
- CRYSTAL92
- CRYSTAL95
- CRYSTAL98
- CRYSTAL03
- CRYSTAL06
- CRYSTAL09
- CRYSTAL14
- CRYSTAL17

Imperial College London

UNIVERSITÀ DI TORINO

CRYSTAL88

CRYSTAL92

CRYSTAL95

CRYSTAL98

CRYSTAL03

CRYSTAL06

CRYSTAL09

CRYSTAL14

CRYSTAL17

A. Erba, J. K. Desmarais, S. Casassa, B. Civalleri, L. Donà, I. J. Bush, B. Searle, L. Maschio, L. Edith-Daga, A. Cossard, C. Ribaldone, E. Ascrizzi, N. L. Marana, J.-P. Flament and B. Kirtman,

J. Chem. Theory Comput., **19**, 6891-6932 (2023)

CRYSTAL23: A Program for Computational Solid State Physics and Chemistry

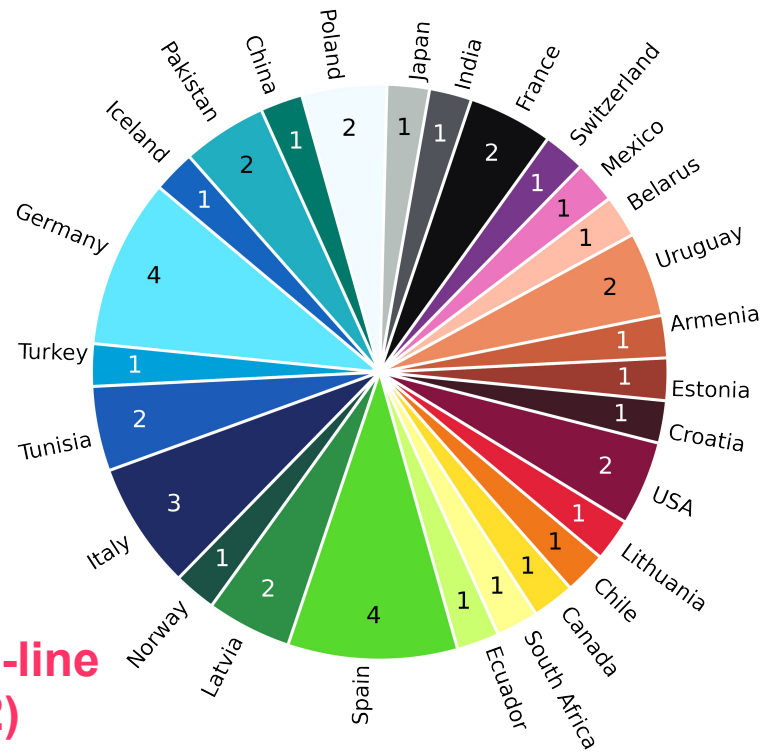
CRYSTAL23

Imperial College London

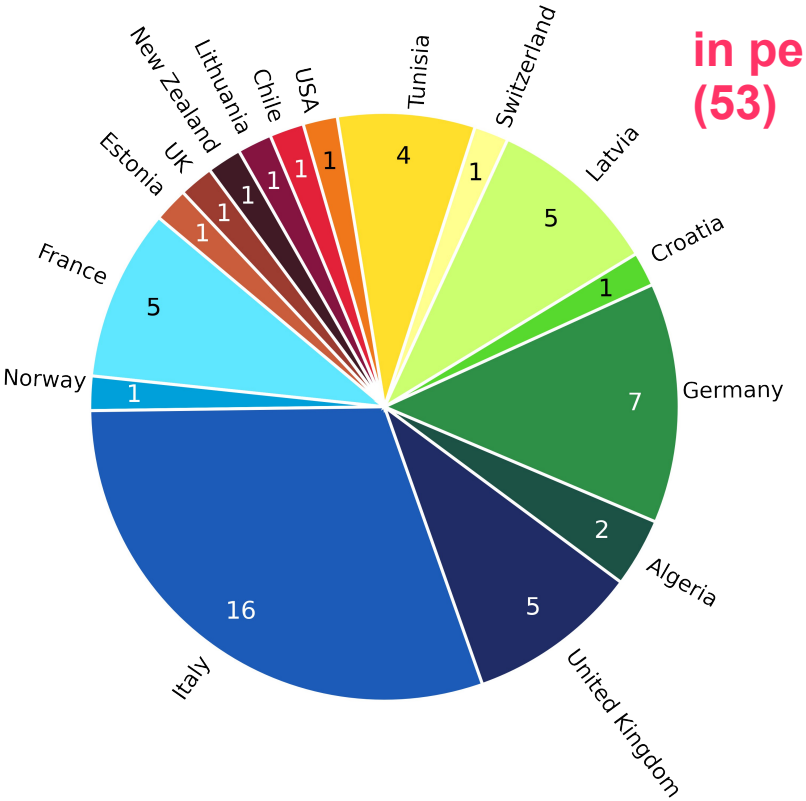
UNIVERSITÀ DI TORINO

MSSC2025

The School runs in hybrid mode.
We have **95** delegates from all over the world (**31** countries):



On-line (42)



in person (53)

VIEW ONLINE

MORNING SESSION

LECTURES

Time	MONDAY 8th	TUESDAY 9th	WEDNESDAY 10th	THURSDAY 11th	FRIDAY 12th
8:15 - 8:40	**Registration**				
8:40 - 9:20	**Opening Remarks** *Alessandro Erba*	**One-Electron Properties of Solids** *Furio Corà*	**Density Functional Perturbation Theory and its Time-Dependent Extension** *Jacques Desmarais* **Calculating Excitons from CRYSTAL** *Manuel Antonio García Blázquez*	**van der Waals Interactions in the DFT Framework** *Jan Gerit Brandenburg*	**Spin-Orbit Coupled Calculations** *Jacques Desmarais*
9:20 - 10:00	**Tackling the Many-Body Problem: the DFT Approach** *Stefano Pittalis*	**Chemical Bonding in Materials** *Jacques Desmarais* **Fock-Exchange in Solids with a Local Basis Set: Hybrid DFAs Made Efficient** *Giacomo Ambrogio*	**Modelling IR and Raman Spectra and Non-Linear Optical Properties** *Lorenzo Maschio*	**Molecular Dynamics with CRYSTAL** *Chiara Ribaldone* **Cost-effective Composite HF/DFT Methods for High Throughput Screening and Machine Learning Applications** *Mattia Raimondo*	**Strongly Anharmonic Systems with CRYSTAL** *Matteo Calandra*
10:00 - 10:25	Cofee Break	Cofee Break	Cofee Break	Cofee Break	Cofee Break
10:25 - 11:05	**Local Gaussian-Type Basis Functions for Solids** *Bartolomeo Civalleri*	**How to Model Defects and Surfaces** *Giuseppe Mallia*	**Density Functional Approach to Magnetism** *Stefano Pittalis*	**Exploiting Parallel Computing with CRYSTAL** *Ian Bush*	**Electron Transport Properties** *Eleonora Ascrizzi* **Modelling Diffusion Coefficients and Ionic Transport** *Laura Bonometti*
11:05 - 11:45	**The Periodic LCAO Approach** *Silvia Casassa*	**Mechanical Response of Materials: Pressure and Elasticity** *Maria Alfredsson*	**Harmonic Lattice Dynamics** *Anna Ferrari*	**Large Scale DFT Simulations on CPU/GPU Architectures** *Giacomo Ambrogio* **How to Model Metal-Organic Frameworks with CRYSTAL** *Lorenzo Donà*	**Phonon Transport Properties** *Antti Karttunen*
11:45 - 12:25	**Treatment of the Coulomb Interaction in Solids** *Jacques Desmarais*	**Predicting Physical Properties of Materials: A Crystallographer Perspective** *Anna Krawczuk*	**Thermodynamics and Thermoelasticity Beyond the Harmonic Approximation** *Alessandro Erba* **Anharmonic Vibrational States and Spectra** *Davide Mitoli*	**Tackling Size and Structural Complexity with CRYSTAL** *Marta Corno*	**Beyond Linear Response: Exploring Nonlinear Optical Phenomena with CRYSTAL** *Panaghiotis Karamanis*
12:25 - 14:30	Lunch	Lunch	Lunch	Lunch	Lunch

MORNING SESSION

LECTURES

Time	MONDAY 8th	TUESDAY 9th	WEDNESDAY 10th	THURSDAY 11th	FRIDAY 12th
8:15 - 8:40	**Registration**				
8:40 - 9:20	**Opening Remarks** *Alessandro Erba*	**One-Electron Properties of Solids** *Furio Corà*	**Density Functional Perturbation Theory and its Time-Dependent Extension** *Jacques Desmarais* **Calculating Excitons from CRYSTAL** *Manuel Antonio García Blázquez*	**van der Waals Interactions in the DFT Framework** *Jan Gerit Brandenburg*	**Spin-Orbit Coupled Calculations** *Jacques Desmarais*
9:20 - 10:00	**Tackling the Many-Body Problem: the DFT Approach** *Stefano Pittalis*	**Chemical Bonding in Materials** *Jacques Desmarais* **Fock-Exchange in Solids with a Local Basis Set: Hybrid DFAs Made Efficient** *Giacomo Ambrogio*	**Modelling IR and Raman Spectra and Non-Linear Optical Properties** *Lorenzo Maschio*	**Molecular Dynamics with CRYSTAL** *Chiara Ribaldone* **Cost-effective Composite HF/DFT Methods for High Throughput Screening and Machine Learning Applications** *Mattia Raimondo*	**Strongly Anharmonic Systems with CRYSTAL** *Matteo Calandra*
10:00 - 10:25	Cofee Break	Cofee Break	Cofee Break	Cofee Break	Cofee Break
10:25 - 11:05	**Local Gaussian-Type Basis Functions for Solids** *Bartolomeo Civalleri*	**How to Model Defects and Surfaces** *Giuseppe Mallia*	**Density Functional Approach to Magnetism** *Stefano Pittalis*	**Exploiting Parallel Computing with CRYSTAL** *Ian Bush*	**Electron Transport Properties** *Eleonora Ascrizzi* **Modelling Diffusion Coefficients and Ionic Transport** *Laura Bonometti*
11:05 - 11:45	**The Periodic LCAO Approach** *Silvia Casassa*	**Mechanical Response of Materials: Pressure and Elasticity** *Maria Alfredsson*	**Harmonic Lattice Dynamics** *Anna Ferrari*	**Large Scale DFT Simulations on CPU/GPU Architectures** *Giacomo Ambrogio* **How to Model Metal-Organic Frameworks with CRYSTAL** *Lorenzo Donà*	**Phonon Transport Properties** *Antti Karttunen*
11:45 - 12:25	**Treatment of the Coulomb Interaction in Solids** *Jacques Desmarais*	**Predicting Physical Properties of Materials: A Crystallographer Perspective** *Anna Krawczuk*	**Thermodynamics and Thermoelasticity Beyond the Harmonic Approximation** *Alessandro Erba* **Anharmonic Vibrational States and Spectra** *Davide Mitoli*	**Tackling Size and Structural Complexity with CRYSTAL** *Marta Corno*	**Beyond Linear Response: Exploring Nonlinear Optical Phenomena with CRYSTAL** *Panaghiotis Karamanis*
12:25 - 14:30	Lunch	Lunch	Lunch	Lunch	Lunch

MSSC2025 Group Photo

VIEW ONLINE

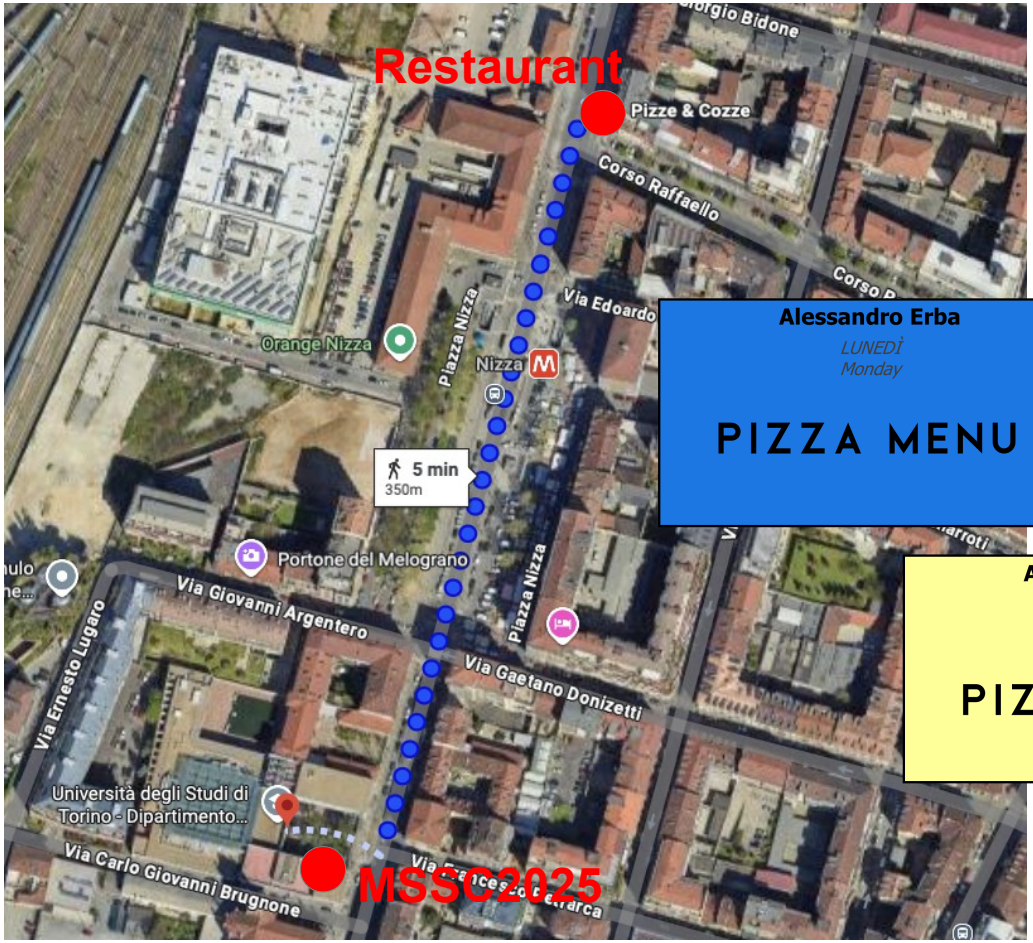
MORNING SESSION

LECTURES

Time	MONDAY 8th	TUESDAY 9th	WEDNESDAY 10th	THURSDAY 11th	FRIDAY 12th
8:15 - 8:40	**Registration**				
8:40 - 9:20	**Opening Remarks** *Alessandro Erba* **Tackling the Many-Body Problem: the DFT Approach** *Stefano Pittalis*	**One-Electron Properties of Solids** *Furio Corà*	**Density Functional Perturbation Theory and its Time-Dependent Extension** *Jacques Desmarais* **Calculating Excitons from CRYSTAL** *Manuel Antonio García Blázquez*	**van der Waals Interactions in the DFT Framework** *Jan Gerit Brandenburg*	**Spin-Orbit Coupled Calculations** *Jacques Desmarais*
9:20 - 10:00		**Chemical Bonding in Materials** *Jacques Desmarais* **Fock-Exchange in Solids with a Local Basis Set: Hybrid DFAs Made Efficient** *Giacomo Ambrogio*	**Modelling IR and Raman Spectra and Non-Linear Optical Properties** *Lorenzo Maschio*	**Molecular Dynamics with CRYSTAL** *Chiara Ribaldone* **Cost-effective Composite HF/DFT Methods for High Throughput Screening and Machine Learning Applications** *Mattia Raimondo*	**Strongly Anharmonic Systems with CRYSTAL** *Matteo Calandra*
10:00 - 10:25	Cofee Break	Cofee Break	Cofee Break	Cofee Break	Cofee Break
10:25 - 11:05	**Local Gaussian-Type Basis Functions for Solids** *Bartolomeo Civalleri*	**How to Model Defects and Surfaces** *Giuseppe Mallia*	**Density Functional Approach to Magnetism** *Stefano Pittalis*	**Exploiting Parallel Computing with CRYSTAL** *Ian Bush*	**Electron Transport Properties** *Eleonora Ascrizzi* **Modelling Diffusion Coefficients and Ionic Transport** *Laura Bonometti*
11:05 - 11:45	**The Periodic LCAO Approach** *Silvia Casassa*	**Mechanical Response of Materials: Pressure and Elasticity** *Maria Alfredsson*	**Harmonic Lattice Dynamics** *Anna Ferrari*	**Large Scale DFT Simulations on CPU/GPU Architectures** *Giacomo Ambrogio* **How to Model Metal-Organic Frameworks with CRYSTAL** *Lorenzo Donà*	**Phonon Transport Properties** *Antti Karttunen*
11:45 - 12:25	**Treatment of the Coulomb Interaction in Solids** *Jacques Desmarais*	**Predicting Physical Properties of Materials: A Crystallographer Perspective** *Anna Krawczuk*	**Thermodynamics and Thermoelasticity Beyond the Harmonic Approximation** *Alessandro Erba* **Anharmonic Vibrational States and Spectra** *Davide Mitoli*	**Tackling Size and Structural Complexity with CRYSTAL** *Marta Corno*	**Beyond Linear Response: Exploring Nonlinear Optical Phenomena with CRYSTAL** *Panagiotis Karamanis*
12:25 - 14:30	Lunch	Lunch	Lunch	Lunch	Lunch

Ristorante Pizze & Cozze
Via Nizza 69

Molecular Biotechnology Center (MBC)
Via Nizza 52



Ristorante Pizze & Cozze
Via Nizza 69

Alessandro Erba
LUNEDÌ
Monday
PIZZA MENU

Alessandro Erba
MARTEDÌ
Tuesday
TRIS MENU
LASAGNA
SPIGOLA IN CROSTA

Alessandro Erba
MERCOLEDÌ
Wednesday
TRIS MENU
RISOTTO GAMBERI E ZUCCHINI
PARMIGIANA

Alessandro Erba
GIOVEDÌ
Thursday
PIZZA MENU

Alessandro Erba
VENERDÌ
Friday
TRIS MENU
RISOTTO ALLE VERDURE
MANZO AFFUMICATO

Molecular Biotechnology Center (MBC)
Via Nizza 52

AFTERNOON SESSION

BEGINNERS

Time	MONDAY 8th	TUESDAY 9th	WEDNESDAY 10th	THURSDAY 11th	FRIDAY 12th
14.30 - 16:00	**Hands-On Kickoff** *Davide Mitoli*	**Geometry Optimization and Equation-of-State** *Giuseppe Mallia*	**Poster Session**	**One-Electron Properties** *Giuseppe Mallia and Silvia Casassa*	**How to Deal with Spin**
	Geometry Input *Andreha Gelli*	**Modelling of 2D Systems** *Giuseppe Mallia*			**Response Properties CPHF/KS** *Chiara Ribaldone*
16:00 - 16:30	Cofee Break	Ice Cream Break	*Free Time*	Ice Cream Break	Cofee Break
16:30 - 18:00	**Basis Set, SCF and Single-Point Energy** *Andreha Gelli*	**Harmonic Frequencies and Spectra** *Davide Mitoli*		**QTAIM** *Marcos Rivera*	**How to Simulate Defects** *Giuseppe Mallia*

EXPERIENCED USERS

Time	MONDAY 8th	TUESDAY 9th	WEDNESDAY 10th	THURSDAY 11th	FRIDAY 12th
14.30 - 16:00	**Hands-On Kickoff**	**Equation-of-State and Pressure**	**Poster Session**	**Harmonic and Anharmonic Vibrational Spectroscopies** *Davide Mitoli*	**Spin-Orbit Coupling and Non-Collinear DFT** *Jacques Desmarais*
	How to Run CRYSTAL in Parallel *Giacomo Ambrogio*	**Elastic Properties** *Filippo Bodo*			
16:00 - 16:30	Cofee Break	Ice Cream Break	*Free Time*	Ice Cream Break	Cofee Break
16:30 - 18:00	**One-Electron Properties** *Laura Bonometti*	**Electron Transport Properties** *Eleonora Ascrizzi*		**Phonons and Thermodynamics** *Giacomo Ambrogio*	**Dispersion Corrections and Composite Methods** *Lorenzo Donà*
	QTAIM *Marcos Rivera*				

AFTERNOON SESSION

MSSC2025
TORINO 8-12 SEP 2025

Molecular Biotechnology Center
Via Nizza 52, 10126 Torino (Italy)

8:40-12:25 **Morning Lectures**
14:30-18:00 **BASIC Tutorial**
Full timetable available online at:
https://mssc.crystalsolutions.eu

unito-guest
User: User
Password: Password

Share your experience with @crystaldevs

NAME
Affiliation

ATTENDEE

IMPERIAL · THOMAS YOUNG CENTRE THE LONDON CENTRE FOR THE THEORY AND SIMULATION OF MATERIALS · UNIVERSITÀ DI TORINO · chimica · 2023 - 2027 DIPARTIMENTO DI ECCELLENZA

Von Neumann Room

MSSC2025
TORINO 8-12 SEP 2025

Molecular Biotechnology Center
Via Nizza 52, 10126 Torino (Italy)

8:40-12:25 **Morning Lectures**
14:30-18:00 **ADVANCED Tutorial**
Full timetable available online at:
https://mssc.crystalsolutions.eu

unito-guest
User: User
Password: Password

Share your experience with @crystaldevs

NAME
Affiliation

ATTENDEE

IMPERIAL · THOMAS YOUNG CENTRE THE LONDON CENTRE FOR THE THEORY AND SIMULATION OF MATERIALS · UNIVERSITÀ DI TORINO · chimica · 2023 - 2027 DIPARTIMENTO DI ECCELLENZA

Copernico Room

AFTERNOON SESSION

BEGINNERS

Time	MONDAY 8th	TUESDAY 9th	WEDNESDAY 10th	THURSDAY 11th	FRIDAY 12th
14.30 - 16:00	**Hands-On Kickoff** *Davide Mitoli*	**Geometry Optimization and Equation-of-State** *Giuseppe Mallia*	**Poster Session**	**One-Electron Properties** *Giuseppe Mallia and Silvia Casassa*	**How to Deal with Spin**
	Geometry Input *Andreha Gelli*	**Modelling of 2D Systems** *Giuseppe Mallia*			**Response Properties CPHF/KS** *Chiara Ribaldone*
16:00 - 16:30	Cofee Break	Ice Cream Break	*Free Time*	Ice Cream Break	Cofee Break
16:30 - 18:00	**Basis Set, SCF and Single-Point Energy** *Andreha Gelli*	**Harmonic Frequencies and Spectra** *Davide Mitoli*		**QTAIM** *Marcos Rivera*	**How to Simulate Defects** *Giuseppe Mallia*

EXPERIENCED USERS

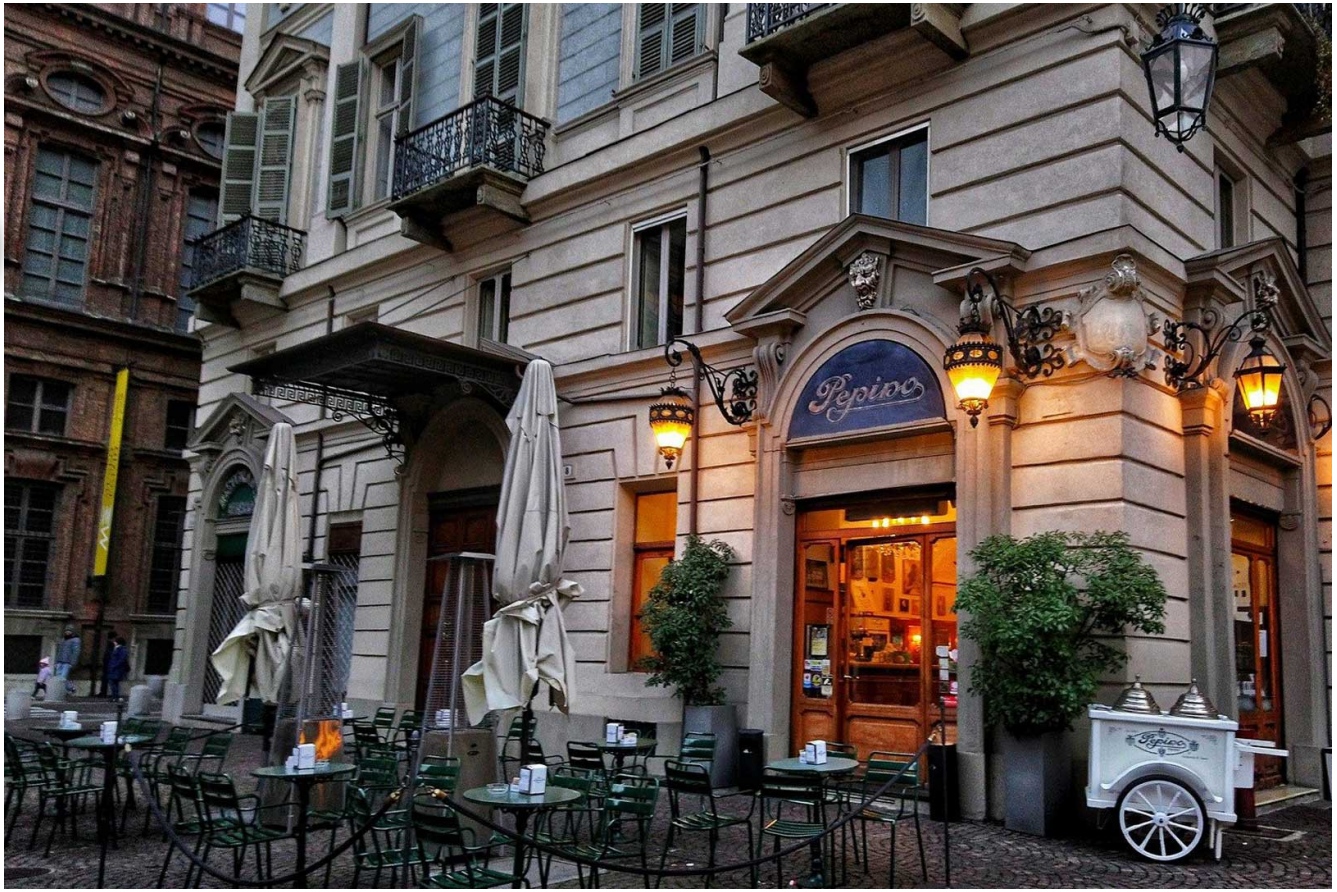
Time	MONDAY 8th	TUESDAY 9th	WEDNESDAY 10th	THURSDAY 11th	FRIDAY 12th
14.30 - 16:00	**Hands-On Kickoff**	**Equation-of-State and Pressure**	**Poster Session**	**Harmonic and Anharmonic Vibrational Spectroscopies** *Davide Mitoli*	**Spin-Orbit Coupling and Non-Collinear DFT** *Jacques Desmarais*
	How to Run CRYSTAL in Parallel *Giacomo Ambrogio*	**Elastic Properties** *Filippo Bodo*			
16:00 - 16:30	Cofee Break	Ice Cream Break	*Free Time*	Ice Cream Break	Cofee Break
16:30 - 18:00	**One-Electron Properties** *Laura Bonometti*	**Electron Transport Properties** *Eleonora Ascrizzi*		**Phonons and Thermodynamics** *Giacomo Ambrogio*	**Dispersion Corrections and Composite Methods** *Lorenzo Donà*
	QTAIM *Marcos Rivera*				

MSSC2025

TORINO 8-12 SEP 2025

VIEW ONLINE

AFTERNOON SESSION



AFTERNOON SESSION

BEGINNERS

Time	MONDAY 8th	TUESDAY 9th	WEDNESDAY 10th	THURSDAY 11th	FRIDAY 12th
14.30 - 16:00	**Hands-On Kickoff** *Davide Mitoli*	**Geometry Optimization and Equation-of-State** *Giuseppe Mallia*	**Poster Session**	**One-Electron Properties** *Giuseppe Mallia and Silvia Casassa*	**How to Deal with Spin**
	Geometry Input *Andreha Gelli*	**Modelling of 2D Systems** *Giuseppe Mallia*			**Response Properties CPHF/KS** *Chiara Ribaldone*
16:00 - 16:30	Cofee Break	Ice Cream Break	*Free Time*	Ice Cream Break	Cofee Break
16:30 - 18:00	**Basis Set, SCF and Single-Point Energy** *Andreha Gelli*	**Harmonic Frequencies and Spectra** *Davide Mitoli*		**QTAIM** *Marcos Rivera*	**How to Simulate Defects** *Giuseppe Mallia*

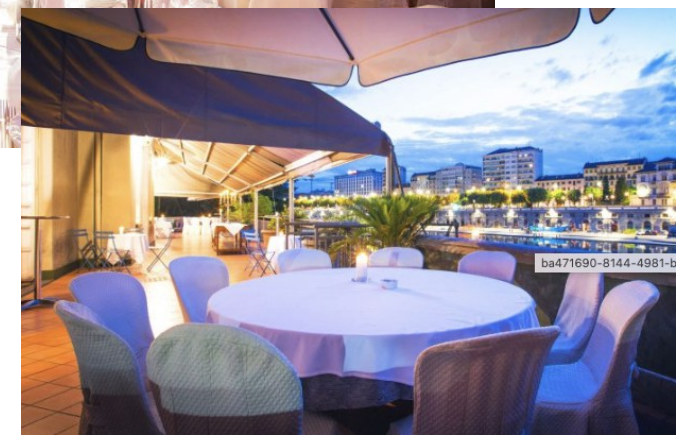
EXPERIENCED USERS

Time	MONDAY 8th	TUESDAY 9th	WEDNESDAY 10th	THURSDAY 11th	FRIDAY 12th
14.30 - 16:00	**Hands-On Kickoff**	**Equation-of-State and Pressure**	**Poster Session**	**Harmonic and Anharmonic Vibrational Spectroscopies** *Davide Mitoli*	**Spin-Orbit Coupling and Non-Collinear DFT** *Jacques Desmarais*
	How to Run CRYSTAL in Parallel *Giacomo Ambrogio*	**Elastic Properties** *Filippo Bodo*			
16:00 - 16:30	Cofee Break	Ice Cream Break	*Free Time*	Ice Cream Break	Cofee Break
16:30 - 18:00	**One-Electron Properties** *Laura Bonometti*	**Electron Transport Properties** *Eleonora Ascrizzi*		**Phonons and Thermodynamics** *Giacomo Ambrogio*	**Dispersion Corrections and Composite Methods** *Lorenzo Donà*
	QTAIM *Marcos Rivera*				

2 Best Poster Awards

Time	MONDAY 8th	TUESDAY 9th	WEDNESDAY 10th	THURSDAY 11th	FRIDAY 12th
19:30 - 23:30			**Social Dinner**		

Esperia Ristorante
C.so Moncalieri 2, Torino



Enjoy MSSC2025!

Instagram

@crystaldevs